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Information processing apparatus, non-transitory computer readable medium storing information processing program, and information processing method

Abstract

An information processing apparatus includes a processor configured to: acquire operation sound data being data obtained by recording an operation sound of a target apparatus used in an organization; detect an abnormal sound from the operation sound data; and transmit partial operation sound data to an out-of-organization processing apparatus, and not transmit a portion of the operation sound data other than the partial operation sound data to the out-of-organization processing apparatus, the partial operation sound data being data that contains the abnormal sound and is a portion of the operation sound data, and the out-of-organization processing apparatus being an apparatus that is managed outside the organization and executes a process related to the abnormal sound.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-040027 filed Mar. 15, 2022.

BACKGROUND

(i) Technical Field

(2) The present invention relates to an information processing apparatus, a non-transitory computer

readable medium storing an information processing program, and an information processing method.

(ii) Related Art

(3) JP4810389B, JP2006-332889A, and JP4706890B disclose an apparatus that detects an occurrence of a failure or a problem in a target apparatus as a target for grasping a situation of the failure or the occurrence of the problem, based on sound recorded data obtained by recording a sound emitted by the target apparatus. Further, JP6868176B discloses a diagnostic apparatus that diagnoses a target apparatus to be diagnosed, based on sound recorded data obtained by recording a sound emitted by the target apparatus, and outputs a warning sound at the start of recording the sound in order to prevent confidential information from being contained in the sound recorded data.

SUMMARY

(4) Meanwhile, a case where operation sound data is acquired by recording an operation sound of a target apparatus, and various processes related to an abnormal sound are executed in a case where the operation sound data contains the abnormal sound is considered. Although not limited to examples as follows, examples of the processes related to the abnormal sound include an abnormality detection process of detecting an abnormality (problem or the like) of a target apparatus based on an abnormal sound, a cause specifying process of specifying the cause of an abnormal sound, and an abnormal sound management process of managing (saving) operation sound data containing an abnormal sound.

(5) Here, in a case where a target apparatus is used in a certain organization (for example, a company), the operation sound of the target apparatus is recorded to acquire operation sound data, and the operation sound data contains an abnormal sound, the operation sound data may be transmitted to an out-of-organization processing apparatus that is managed outside the organization, and the out-of-organization processing apparatus executes the process related to the abnormal sound. In such a case, in a case where the operation sound data contains a sound representing the confidential information of the organization, the confidential information is leaked to the outside of the organization. Thus, in such a case, it is required to suppress the leakage of the confidential information. The confidential information refers to information that does not open to the outside of the organization. Examples of the confidential information include confidential information possessed by the organization and privacy information regarding a person in the organization.

(6) In addition, JP6868176B discloses that the warning sound is output at the start of recording a sound so that the sound recorded data does not contain the confidential information. However, according to JP6868176B, the warning sound is output every time the operation sound of the target apparatus is recorded, and this may be considerably troublesome. Therefore, it is required to suppress the leakage of the confidential information by another method.

(7) Aspects of non-limiting embodiments of the present disclosure relate to an information processing apparatus, a non-transitory computer readable medium storing an information processing program, and an information processing method that suppress leakage of confidential information of an organization in a case where an out-of-organization processing apparatus executes a process related to an abnormal sound contained in operation sound data obtained by recording a sound from an operation of a target apparatus used in the organization, as compared to a case where the entirety of the operation sound data is transmitted to the out-of-organization processing apparatus.

(8) Aspects of certain non-limiting embodiments of the present disclosure overcome the above disadvantages and/or other disadvantages not described above. However, aspects of the non-limiting embodiments are not required to overcome the disadvantages described above, and aspects of the non-limiting embodiments of the present disclosure may not overcome any of the disadvantages described above.

(9) According to an aspect of the present disclosure, there is provided an information processing

apparatus including a processor configured to: acquire operation sound data being data obtained by recording an operation sound of a target apparatus used in an organization; detect an abnormal sound from the operation sound data; and transmit partial operation sound data to an out-of-organization processing apparatus, and not transmit a portion of the operation sound data other than the partial operation sound data to the out-of-organization processing apparatus, the partial operation sound data being data that contains the abnormal sound and is a portion of the operation sound data, and the out-of-organization processing apparatus being an apparatus that is managed outside the organization and executes a process related to the abnormal sound.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:
- (2) FIG. 1 is a schematic diagram illustrating a configuration of an information processing system according to the present exemplary embodiment;
- (3) FIG. 2 is a schematic diagram illustrating a configuration of a target apparatus according to the present exemplary embodiment;
- (4) FIG. 3 is a diagram illustrating an example of operation sound data;
- (5) FIG. 4 is a diagram illustrating a first example of a content of a notification transmitted to a data checker;
- (6) FIG. 5 is a diagram illustrating a second example of the content of the notification transmitted to the data checker;
- (7) FIG. 6 is a diagram illustrating a third example of the content of the notification transmitted to the data checker;
- (8) FIG. 7 is a first flowchart illustrating a processing flow of an information processing apparatus according to the present exemplary embodiment; and
- (9) FIG. 8 is a second flowchart illustrating the processing flow of the information processing apparatus according to the present exemplary embodiment.

DETAILED DESCRIPTION

(10) FIG. 1 is a schematic diagram illustrating a configuration of an information processing system **10** according to the present exemplary embodiment. The information processing system **10** includes a target apparatus **12** as an information processing apparatus, a checker terminal **14**, and an out-of-organization processing apparatus **16**. The target apparatus **12** and the checker terminal **14** are communicably connected to each other by an in-organization communication line **18** such as a local area network (LAN). The target apparatus **12** and the out-of-organization processing apparatus **16** are communicably connected to each other by a public communication line **20** such as a wide area network (WAN). Further, the checker terminal **14** and the out-of-organization processing apparatus **16** are also communicably connected to each other by the public communication line **20**.

(11) The target apparatus **12** is an apparatus used in a certain organization. In other words, the target apparatus **12** is installed in a certain organization. Here, the organization refers to an organization such as a company, a research institute, a government office, or a university. A portion of the organization (for example, one department or section in a company) may be provided as one organization. In particular, the organization here means an organization having confidential information. As described above, the confidential information refers to information that does not open to the outside of the organization. Examples of the confidential information include confidential information possessed by the organization and privacy information regarding a person in the organization.

(12) The target apparatus **12** in the present exemplary embodiment is a multifunction device, a printer, or the like, but the target apparatus **12** may be any apparatus as long as the apparatus emits an operation sound. The details of the target apparatus **12** will be described later.

(13) The checker terminal **14** is a terminal used by a data checker who checks whether or not confidential information of the organization is contained in partial operation sound data (details will be described later) that contains an abnormal sound and is a portion of operation sound data obtained by recording an operation sound of the target apparatus **12**. The data checker is a person in the organization in which the target apparatus **12** is installed. For example, the data checker may be a general user who uses the target apparatus **12**, or may be an administrator who manages the target apparatus **12**.

(14) The checker terminal **14** is configured by, for example, a computer such as a personal computer, a tablet terminal, or a smartphone, but is not limited to the above devices. The checker terminal **14** is configured to include a communication interface, an input interface, a display, a memory, a processor, and the like. The communication interface is provided to communicate with the target apparatus **12** through the in-organization communication line **18** or to communicate with the out-of-organization processing apparatus **16** through the public communication line **20**. The input interface is provided to receive a command of the data checker. The display displays a screen or the like provided from the target apparatus **12**. The memory is provided to store data. The processor executes each process.

(15) The out-of-organization processing apparatus **16** is an apparatus that executes a process related to an abnormal sound contained in the above-described partial operation sound data. Examples of the process related to the abnormal sound, which is executed by the out-of-organization processing apparatus **16**, include an abnormality detection process of detecting an abnormality (problem or the like) of the target apparatus **12** based on the abnormal sound, a cause specifying process of specifying the cause of an abnormal sound, and an abnormal sound management process of managing (saving) operation sound data containing an abnormal sound. Such a process is not limited to the above examples.

(16) The out-of-organization processing apparatus **16** is an apparatus managed outside the organization in which the target apparatus **12** is installed. For example, in a case where the target apparatus **12** is a multifunction device or a printer, the out-of-organization processing apparatus **16** may be a server managed by the manufacturer of the target apparatus **12**. The out-of-organization processing apparatus **16** is not limited to this. The out-of-organization processing apparatus **16** is configured to include a communication interface for communicating with the target apparatus **12** or the checker terminal **14** through the public communication line **20**, a memory for storing partial operation sound data and the like, a processor that executes a process related to an abnormal sound contained in the partial operation sound data, and the like.

(17) Since the out-of-organization processing apparatus **16** is an apparatus managed outside the organization, the partial operation sound data transmitted to the out-of-organization processing apparatus **16** may be heard by a person outside the organization. For example, in a case where the partial operation sound data contains confidential information (which may also be called a confidential voice) of the organization, such as a voice in a meeting in the organization, the conversation of people in the organization, and other sounds generated in the organization, this confidential information will be leaked outside the organization. In the information processing system **10** according to the present exemplary embodiment, as will be described later, leakage of the confidential information to the outside of the organization is suppressed while causing the out-of-organization processing apparatus **16** to execute the process related to the partial operation sound data of the target apparatus **12**.

(18) FIG. 2 is a schematic diagram illustrating the configuration of the target apparatus **12**.

(19) A communication interface **30** is configured to include, for example, a network interface card (NIC). The communication interface **30** exhibits a function of communicating with the checker

terminal **14** through the in-organization communication line **18**. In addition, the communication interface **30** exhibits a function of communicating with the out-of-organization processing apparatus **16** through the public communication line **20**.

(20) An input interface **32** is configured to include, for example, various buttons and a touch panel. The input interface **32** is used to input an instruction from a general user of the target apparatus **12** to the target apparatus **12**.

(21) A display **34** includes, for example, a liquid crystal panel and an organic electro luminescence (EL) panel. Various screens are displayed on the display **34**.

(22) A microphone **36** is installed near the target apparatus **12** and converts an operation sound of the target apparatus **12** into operation sound data which is an electric signal.

(23) A volatile memory **38** includes a random access memory (RAM) and the like. The volatile memory **38** is a memory in which storing data is not possible unless power is supplied. In other words, in a case where the power of the target apparatus **12** is turned off and the power supply to the volatile memory **38** is cut off, data stored in the volatile memory **38** is erased. The volatile memory **38** stores one or a plurality of pieces of operation sound data **40** of the target apparatus **12**, which have been acquired by the microphone **36**. The operation sound data **40** contains partial operation sound data **42** which is a portion of the operation sound data **40** and contains an abnormal sound. The partial operation sound data **42** is cut out from the operation sound data **40** by a processor **46** (specifically, partial operation sound data acquisition unit **52**).

(24) A non-volatile memory **44** is configured to include a hard disk drive (HDD), a solid state drive (SSD), an embedded multimedia card (eMMC), a read only memory (ROM), and the like. The non-volatile memory **44** is a memory that can store data without supplying the power. In other words, even though the power of the target apparatus **12** is turned off and the power supply to the non-volatile memory **44** is cut off, the data stored in the non-volatile memory **44** is not erased. The non-volatile memory **44** stores an information processing program for operating each unit of the target apparatus **12**. The information processing program can also be stored in a non-transitory computer readable storage medium such as a universal serial bus (USB) memory or a CD-ROM. The target apparatus **12** can read and execute the information processing program from such a storage medium.

(25) In the embodiments above, the term “processor” refers to hardware in a broad sense. Examples of the processor include general processors (e.g., CPU: Central Processing Unit) and dedicated processors (e.g., GPU: Graphics Processing Unit, ASIC: Application Specific Integrated Circuit, FPGA: Field Programmable Gate Array, and programmable logic device). In the embodiments above, the term “processor” is broad enough to encompass one processor or plural processors in collaboration which are located physically apart from each other but may work cooperatively. The order of operations of the processor is not limited to one described in the embodiments above, and may be changed.

(26) As illustrated in FIG. 2, the processor **46** exhibits functions as an operation sound data acquisition processing unit **48**, an abnormal sound detection unit **50**, a partial operation sound data acquisition unit **52**, and a transmission processing unit **54**, in accordance with the information processing program stored in the non-volatile memory **44**. First, the processor **46** authenticates a general user by, for example, a method of reading an ID card of the user. Then, by receiving a process request (for example, a print job) from the general user, the processor **46** executes a process related to the process request. Simultaneously, the operation sound data acquisition processing unit **48**, the abnormal sound detection unit **50**, the partial operation sound data acquisition unit **52**, and the transmission processing unit **54** sequentially execute processes.

(27) In a case where the target apparatus **12** starts an operation (in a case where the target apparatus **12** starts the process related to the process request in the present exemplary embodiment), the operation sound data acquisition processing unit **48** records the operation sound of the target apparatus **12** with the microphone **36** to acquire operation sound data **40**. The operation sound data

acquisition processing unit **48** stores the acquired operation sound data **40** in the volatile memory **38**. As described above, in the present exemplary embodiment, the operation sound data acquisition processing unit **48** acquires the operation sound data **40** and stores the acquired operation sound data **40** in the volatile memory **38**, every time the target apparatus **12** operates. FIG. **3** illustrates an example of the operation sound data **40**.

(28) The abnormal sound detection unit **50** detects an abnormal sound N from the operation sound data **40**. Various methods can be adopted as a method of detecting the abnormal sound N. For example, the abnormal sound N can be detected in a manner that normal operation sound data in a case where the target apparatus **12** normally operates is acquired in advance and is stored in the non-volatile memory **44**, and the operation sound data **40** is compared to the normal operation sound data. Alternatively, the abnormal sound N may be detected by inputting the operation sound data **40** into a learning device that has been sufficiently learned to detect the abnormal sound from the operation sound data **40**.

(29) The abnormal sound detection unit **50** detects not only whether or not the operation sound data **40** contains the abnormal sound N, but also an occurrence timing (occurrence time point) of the abnormal sound N. The occurrence timing of the abnormal sound N can be represented by, for example, a time from the start time point of the operation sound data **40**.

(30) There are various types of abnormal sounds N contained in the operation sound data **40** in accordance with a form of the occurrence of the abnormal sound. For example, there are a periodic abnormal sound that occurs periodically, such as a plurality of abnormal sounds N illustrated in FIG. **3**, an irregular abnormal sound that occurs irregularly, a continuous abnormal sound that occurs continuously, a single-shot abnormal sound that occurs in a single shot, and the like. The abnormal sound detection unit **50** can specify the type of the abnormal sound N contained in the operation sound data **40** based on the number of abnormal sounds N contained in the operation sound data **40**, the occurrence time interval of the abnormal sound N, and the like.

(31) For example, the abnormal sound detection unit **50** may detect the abnormal sound N from the operation sound data **40**, and simultaneously (in other words, in real time), the operation sound data acquisition processing unit **48** may acquire the operation sound data **40**.

(32) The partial operation sound data acquisition unit **52** executes a process of cutting out the partial operation sound data **42**, which is a portion of the operation sound data **40** and contains the abnormal sound N detected by the abnormal sound detection unit **50**, from the operation sound data **40**. In the present exemplary embodiment, the partial operation sound data acquisition unit **52** acquires a portion having a predetermined time length T from the occurrence timing of the abnormal sound N, as the partial operation sound data **42**. For example, in the example in FIG. **3**, a portion of the time length T from the occurrence timing t1 of the first abnormal sound N contained in the operation sound data **40** is acquired as the partial operation sound data **42**.

(33) The time length T of the partial operation sound data **42** may be predetermined by the administrator of the target apparatus **12** or the like. From the viewpoint of suppressing the leakage of confidential information of the organization, for example, the time length T is required to be as short as possible. However, in a case where the time length T is too short, it may not be possible that the out-of-organization processing apparatus **16** executes the cause specifying process of specifying the cause of the abnormal sound N. Thus, the time length T may be, for example, a time required to specify the cause of the abnormal sound N contained in the partial operation sound data **42** (for example, more preferably a necessary and sufficient time).

(34) The time required to specify the cause of the abnormal sound N contained in the partial operation sound data **42** may differ depending on the type of the abnormal sound contained in the partial operation sound data **42**. For example, for the periodic abnormal sound, it is appropriate that the abnormal sound N for a plurality of cycles is contained in the partial operation sound data **42**. Thus, the time length T may be set to a time including, for example, a plurality of cycles. For example, in a case where the periodic abnormal sound occurs at 1-second intervals, the time length

T can be set to about 8 seconds (8 cycles). On the other hand, since there is no periodicity for the irregular abnormal sound, the continuous abnormal sound, and the single-shot abnormal sound, the time length T can be set to a time shorter than the time length T for the periodic abnormal sound, for example, set to about 1 second. As described above, the time length T of the partial operation sound data **42** may be, for example, a time corresponding to the type of the abnormal sound N contained in the operation sound data **40**.

(35) A time length table in which the type of abnormal sound N contained in the operation sound data **40** and the time length T of the partial operation sound data **42** are associated with each other is prepared in advance and stored in the non-volatile memory **44**. The partial operation sound data acquisition unit **52** specifies the time length T based on the time length table and the type of the abnormal sound N contained in the operation sound data **40** specified by the abnormal sound detection unit **50**. The partial operation sound data acquisition unit **52** acquires the partial operation sound data **42** having the specified time length T.

(36) There is also considered a case where it is not possible to acquire the partial operation sound data **42** having the predetermined time length T, for example, a case where the time length of the operation sound data **40** is short. In this case, the partial operation sound data acquisition unit **52** may use, as the partial operation sound data **42**, a portion from the occurrence timing of the abnormal sound N to the end of the operation sound data **40**.

(37) The transmission processing unit **54** transmits the partial operation sound data **42** acquired by the partial operation sound data acquisition unit **52** to the data checker. As described above, the data checker may be a general user who has input a process request to the target apparatus **12**, or may be an administrator who manages the target apparatus **12**. The data checker checks whether the partial operation sound data **42** contains the confidential information of the organization, and then transmits the partial operation sound data **42** to the out-of-organization processing apparatus **16**. On the other hand, the transmission processing unit **54** does not transmit a portion of the operation sound data **40** acquired by the operation sound data acquisition processing unit **48**, other than the partial operation sound data **42** to the data checker. That is, in the present exemplary embodiment, the portion of the operation sound data **40** other than the partial operation sound data **42** does not leak to the target apparatus **12**.

(38) One aspect of transmitting the partial operation sound data **42** to the data checker is to transmit the partial operation sound data **42** to the checker terminal **14** used by the data checker. The transmission processing unit **54** can transmit a notification including the partial operation sound data **42** to the checker terminal **14**. In the present exemplary embodiment, an e-mail having the partial operation sound data **42** as an attached file is transmitted to the data checker as the notification. Such a notification may have a form other than the e-mail.

(39) The data checker receives the e-mail on the checker terminal **14**, and thereby the checker terminal **14** receives the partial operation sound data **42**. In a case where the data checker is a general user, a user DB (database) in which a user ID for uniquely identifying the general user and an e-mail address of the user are associated with each other can be stored in a server or the like accessible from the target apparatus **12**. The transmission processing unit **54** can acquire the e-mail address of the general user based on the user DB and the user ID obtained in a case where the general user has been authenticated. Even though the data checker is an administrator, an e-mail address of the administrator can be registered in a server or the like accessible from the target apparatus **12**. The transmission processing unit **54** accesses the server or the like to acquire the e-mail address of the administrator.

(40) FIG. **4** is a diagram illustrating a first example of the content of the notification transmitted to the data checker. The body of the e-mail as the notification illustrated in FIG. **4** includes a message indicating that an abnormal sound N has occurred in the target apparatus **12**, a required to check whether or not that the attached partial operation sound data **42** contains confidential information of the organization, a required to transmit the partial operation sound data **42** to the out-of-

organization processing apparatus **16** in a case where the partial operation sound data **42** does not contain the confidential information of the organization, and a transmission destination of the partial operation sound data **42** (e-mail address of the out-of-organization processing apparatus **16**, a uniform resource locator (URL) of a registration destination, or the like). That is, the notification transmitted to the data checker can be information for transmitting the partial operation sound data **42** to the out-of-organization processing apparatus **16** in a state where the data checker is caused to check the partial operation sound data **42**.

(41) The data checker listens to the attached partial operation sound data **42** in accordance with the e-mail transmitted from the transmission processing unit **54**, and checks whether or not the partial operation sound data **42** contains the confidential information of the organization. In a case where the partial operation sound data **42** does not contain the confidential information of the organization, the partial operation sound data **42** is transmitted from the checker terminal **14** to the transmission destination (that is, the out-of-organization processing apparatus **16**) described in the e-mail.

(42) The out-of-organization processing apparatus **16** that has received the partial operation sound data **42** executes a process related to the abnormal sound N contained in the partial operation sound data **42**. A serviceman or the like who performs maintenance of the target apparatus **12** is notified of the result of the process.

(43) Another aspect of transmitting the partial operation sound data **42** to the data checker is to notify the data checker of a link destination (storage location) of the partial operation sound data **42** and to cause the data manager to access the link destination and acquire (download) the partial operation sound data **42**. For example, the transmission processing unit **54** transmits a notification (for example, e-mail) indicating the link destination for the partial operation sound data **42** stored in the volatile memory **38**, to the data checker. Alternatively, the transmission processing unit **54** transmits and stores the partial operation sound data **42** to an in-organization server managed in the organization, and then transmits, to the data checker, the information indicating the link destination of the partial operation sound data **42** stored in the in-organization server. The data checker downloads the partial operation sound data **42** to the checker terminal **14** based on the information indicating the link destination.

(44) In the above-described exemplary embodiment, the transmission processing unit **54** transmits the partial operation sound data **42** to the data checker, and the data checker transmits the partial operation sound data **42** to the out-of-organization processing apparatus **16**. The transmission processing unit **54** may directly transmit the partial operation sound data **42** to the out-of-organization processing apparatus **16**. Also in this case, the transmission processing unit **54** does not transmit the portion of the operation sound data **40** acquired by the operation sound data acquisition processing unit **48**, other than the partial operation sound data **42**, to the out-of-organization processing apparatus **16**.

(45) Further, the transmission processing unit **54** may transmit the partial operation sound data **42** to the data checker, and transmit the partial operation sound data **42** to the out-of-organization processing apparatus **16** in response to the notification from the data checker who has checked that the partial operation sound data **42** does not contain the confidential information of the organization. For example, the transmission processing unit **54** transmits the notification of the content as illustrated in FIG. 5 to the data checker. The notification contains the occurrence of the abnormal sound N in the target apparatus **12**, a request to check whether the attached partial operation sound data **42** contains the confidential information of the organization, a response request in a case where the partial operation sound data **42** does not contain the confidential information of the organization, and a link destination of the partial operation sound data **42**. Meanwhile, "YYaaa.bb.com" in the link destination indicates a folder path in the target apparatus **12**. An e-mail having the attached partial operation sound data **42** may be transmitted to the data checker as the notification.

(46) The data checker downloads the partial operation sound data **42** to the checker terminal **14** from the link destination contained in the notification, and checks the partial operation sound data **42**. As a result, in a case where the data checker determines that the partial operation sound data **42** does not contain the confidential information of the organization, the data checker operates a transmit button contained in the notification. In a case where the transmit button is operated, a check notification is transmitted from the checker terminal **14** to the target apparatus **12**. The transmission processing unit **54** transmits the partial operation sound data **42** to the out-of-organization processing apparatus **16** in response to reception of the check notification. On the other hand, in a case where the data checker determines that the partial operation sound data **42** contains the confidential information of the organization, the data checker operates a cancel button contained in the notification. In a case where the cancel button is operated, a cancellation notification is transmitted from the checker terminal **14** to the target apparatus **12**. In a case where the transmission processing unit **54** receives the cancellation notification, the transmission processing unit **54** does not transmit the partial operation sound data **42** to the out-of-organization processing apparatus **16**.

(47) As described above, in the present exemplary embodiment, the transmission processing unit **54** transmits not the entirety of the operation sound data **40** but only the partial operation sound data **42** being a portion of the operation sound data **40** to the data checker or the out-of-organization processing apparatus **16**. Thus, as compared with at least a case where the entirety of the operation sound data **40** is transmitted to the data checker or the out-of-organization processing apparatus **16**, the occurrence of a situation in which the confidential information of the organization which is contained in the operation sound data **40** is leaked to the outside of the organization is suppressed. For example, even though the portion of the operation sound data **40** other than the partial operation sound data **42** contains the confidential information of the organization, in a case where the partial operation sound data **42** does not contain the confidential information of the organization, the confidential information is not leaked to the outside of the organization. In particular, by the data checker checking the partial operation sound data **42** before the partial operation sound data **42** is transmitted to the out-of-organization processing apparatus **16**, the occurrence of the situation in which the confidential information of the organization which is contained in the operation sound data **40** is leaked to the outside of the organization is more suppressed.

(48) Further, since the operation sound data **40** is stored in the volatile memory **38**, the operation sound data **40** is erased in a case where the power of the target apparatus **12** is turned off. Thus, an occurrence of a situation in which the operation sound data **40** is leaked to the outside of the target apparatus **12**, in particular, to the outside of the organization is suppressed.

(49) As described above, in the present exemplary embodiment, the operation sound data acquisition processing unit **48** acquires the operation sound data **40** every time the target apparatus **12** operates. Thus, in a case where the target apparatus **12** has a problem, a notification is transmitted to the data checker (general user or administrator) every time the target apparatus **12** operates. In a case where the notification is transmitted, particularly for the administrator, it is required to check the partial operation sound data **42** every time receiving the notification, and this may take the large time and effort.

(50) Considering this point, in a case where the partial operation sound data acquisition unit **52** accumulates and stores the acquired partial operation sound data **42** in the volatile memory **38**, and the transmission processing unit **54** satisfies a predetermined transmission condition, for example, in a state where a priority is added to a plurality of pieces of partial operation sound data **42** stored in the volatile memory **38**, the plurality of pieces of partial operation sound data **42** having the priority may be transmitted to the data checker.

(51) Even in this case, the above description is similar to the above exemplary embodiment in that, every time the target apparatus **12** operates, the operation sound data acquisition processing unit **48**

acquires operation sound data **40** and stores the operation sound data **40** in the volatile memory **38**, the abnormal sound detection unit **50** detects an abnormal sound N from the operation sound data **40**, and the partial operation sound data acquisition unit **52** acquires partial operation sound data **42** containing the abnormal sound N, from the operation sound data **40**. However, the transmission processing unit **54** does not immediately transmit the partial operation sound data **42** to the data checker, and stores the partial operation sound data **42** in the volatile memory **38** until the transmission condition is satisfied. Thus, until the transmission condition is satisfied, the partial operation sound data acquisition unit **52** may acquire a plurality of pieces of partial operation sound data **42**, and the plurality of pieces of partial operation sound data **42** may be stored in the volatile memory **38**.

(52) The transmission condition may be predetermined by the administrator of the target apparatus **12** or the like. Examples of the transmission conditions include a case where the time has reached a time point designated in advance, or a case where a predetermined amount (number or data volume) of partial operation sound data **42** is stored in the volatile memory **38**, but the present exemplary embodiment is not limited to this.

(53) Further, in a case where the transmission condition is satisfied, the transmission processing unit **54** adds a priority to a plurality of pieces of partial operation sound data **42** stored in the volatile memory **38**. The priority here indicates the order in which the data checker is to check the data. Specifically, the transmission processing unit **54** adds the priority to each piece of partial operation sound data **42** based on at least one of a sound volume of the abnormal sound N contained in the partial operation sound data **42**, a data volume of the partial operation sound data **42**, or the occurrence time point of the abnormal sound N.

(54) The sound volume of the abnormal sound N may be a peak-to-peak sound pressure level P of the abnormal sound N in a sound wave form (see FIG. 3) of the partial operation sound data **42**. Generally, the larger the sound volume (sound pressure level P) of the abnormal sound N, the easier it is to execute the process related to the abnormal sound N. Thus, the larger sound volume of the abnormal sound N acts in a direction of higher priority.

(55) In the present exemplary embodiment, the operation sound data **40** or the partial operation sound data **42** is in a predetermined file format (for example, a wav format). Therefore, the data volume of the partial operation sound data **42** may indicate the time length T of the partial operation sound data **42**. In general, the longer the time length T, the easier it is to execute the process related to the abnormal sound N. Thus, the longer time length T acts in the direction of higher priority.

(56) Further, in many cases, it is intended to give priority to a process related to the abnormal sound N that has occurred earlier in time. Therefore, a case where the occurrence time point of the abnormal sound N is earlier in time acts in the direction of the higher priority.

(57) The transmission processing unit **54** may add the priority to each piece of partial operation sound data **42** in consideration of at least two of the sound volume of the abnormal sound N contained in the partial operation sound data **42**, the data volume of the partial operation sound data **42**, or the occurrence time point of the abnormal sound N. In this case, the priority of each piece of partial operation sound data **42** may be specified in consideration of the weight added to each of parameters of the sound volume of the abnormal sound N, the data volume of the partial operation sound data **42**, and the occurrence time point of the abnormal sound N.

(58) The transmission processing unit **54** transmits the plurality of pieces of partial operation sound data **42** with the priority to the data checker. For example, the transmission processing unit **54** transmits the notification of the content as illustrated in FIG. 6 to the data checker. The notification includes the occurrence of the abnormal sound N in the target apparatus **12**, a request to check whether or not the attached partial operation sound data **42** contains the confidential information of the organization, a request to transmit the partial operation sound data **42** to the out-of-organization processing apparatus **16** in a case where the partial operation sound data **42** does not contain the

confidential information of the organization, the priorities and link destinations of a plurality of pieces of partial operation sound data **42**, and information indicating that one piece of partial operation sound data **42** may be transmitted to the out-of-organization processing apparatus **16**. A plurality of pieces of partial operation sound data **42** may also be attached to the notification, and this notification may be transmitted to the checker terminal **14**.

(59) The data checker checks whether or not the partial operation sound data **42** contains the confidential information of the organization, based on this notification in order indicated by the priority. In a case where the data checker checks that the partial operation sound data **42** does not contain the confidential information, it is not necessary for the data checker to check the partial operation sound data **42** having a priority lower than the priority of such partial operation sound data that does not contain the confidential information. That is, the time and effort of the data checker is reduced as compared with a case where the partial operation sound data **42** is individually transmitted. The data checker transmits the partial operation sound data **42** checked not to contain the confidential information, to the out-of-organization processing apparatus **16**.

(60) Further, even in the exemplary embodiment in which the partial operation sound data **42** is directly transmitted from the target apparatus **12** to the out-of-organization processing apparatus **16**, in a case where a problem has occurred in the target apparatus **12**, the partial operation sound data **42** is transmitted to the out-of-organization processing apparatus **16** every time the target apparatus **12** operates. In a case where the partial operation sound data **42** is transmitted to the out-of-organization processing apparatus **16** every time the target apparatus **12** operates, a probability that the partial operation sound data **42** containing the confidential information of the organization is transmitted to the out-of-organization processing apparatus **16** may increase.

(61) Considering this point, in a case where the partial operation sound data acquisition unit **52** accumulates and stores the acquired partial operation sound data **42** in the volatile memory **38**, and the transmission processing unit **54** satisfies a predetermined transmission condition, for example, the priority may be added to the plurality of pieces of partial operation sound data **42** stored in the volatile memory **38**, and the partial operation sound data **42** selected in accordance with the priority among the plurality of pieces of partial operation sound data **42** may be transmitted to the out-of-organization processing apparatus **16**.

(62) Specifically, every time the target apparatus **12** operates, the operation sound data acquisition processing unit **48** acquires the operation sound data **40** and stores the operation sound data **40** in the volatile memory **38**, the abnormal sound detection unit **50** detects the abnormal sound N from the operation sound data **40**, and the partial operation sound data acquisition unit **52** acquires the partial operation sound data **42** containing the abnormal sound N, from the operation sound data **40**. The transmission processing unit **54** does not immediately transmit the partial operation sound data **42** to the data checker, and stores the partial operation sound data **42** in the volatile memory **38** until the transmission condition is satisfied. Thus, until the transmission condition is satisfied, the partial operation sound data acquisition unit **52** may acquire a plurality of pieces of partial operation sound data **42**, and the plurality of pieces of partial operation sound data **42** may be stored in the volatile memory **38**.

(63) In a case where the transmission condition is satisfied, as described above, the transmission processing unit **54** adds the priority to each piece of partial operation sound data **42** based on at least one of the sound volume of the abnormal sound N contained in the partial operation sound data **42**, the data volume of the partial operation sound data **42**, or the occurrence time point of the abnormal sound N. The priority here indicates the order in which the transmission processing unit **54** is to perform transmission to the out-of-organization processing apparatus **16**. The transmission processing unit **54** transmits, to the out-of-organization processing apparatus **16**, the partial operation sound data **42** selected based on the priority added to each piece of partial operation sound data **42** among the plurality of pieces of partial operation sound data **42** stored in the volatile memory **38**. Typically, the transmission processing unit **54** transmits only the partial operation

sound data **42** having the highest priority to the out-of-organization processing apparatus **16**.

(64) Even in this case, the transmission processing unit **54** may transmit the plurality of pieces of partial operation sound data **42** with the priority to the data checker, and may transmit the instructed partial operation sound data **42** to the out-of-organization processing apparatus **16** in response to a notification indicating the partial operation sound data **42** to be transmitted to the out-of-organization processing apparatus **16**, which is transmitted from the data checker who has checked that one or the plurality of pieces of partial operation sound data **42** do not contain the confidential information of the organization, in accordance with the priority added to each piece of partial operation sound data **42**.

(65) The configuration of the information processing system **10** according to the present exemplary embodiment is as described above. The processing flow of the target apparatus **12** according to the present exemplary embodiment will be described below with reference to the flowchart illustrated in FIGS. **7** and **8**.

(66) FIG. **7** is a flowchart illustrating a processing flow of the target apparatus **12** in the exemplary embodiment in which the transmission processing unit **54** transmits the partial operation sound data **42** to the data checker, and the data checker transmits the partial operation sound data **42** to the out-of-organization processing apparatus **16**.

(67) In Step **S10**, in a case where the processor **46** receives a process request from a general user, and starts a process related to the received process request, the operation sound data acquisition processing unit **48** records an operation sound of the target apparatus **12** to acquire operation sound data **40** and stores the operation sound data **40** in the volatile memory **38**.

(68) In Step **S12**, the abnormal sound detection unit **50** determines whether or not an abnormal sound **N** is contained in the operation sound data **40** acquired in Step **S10**. In a case where the abnormal sound **N** is detected from the operation sound data **40**, the process proceeds to Step **S14**. In a case where the abnormal sound **N** is not detected, the process bypasses Step **S14** and proceeds to Step **S16**.

(69) In Step **S14**, the partial operation sound data acquisition unit **52** acquires partial operation sound data **42** by cutting out the partial operation sound data **42** containing the abnormal sound **N** detected in Step **S12**, from the operation sound data **40**. Then, the partial operation sound data acquisition unit **52** stores the acquired partial operation sound data **42** in the volatile memory **38**.

(70) In Step **S16**, the transmission processing unit **54** determines whether or not the transmission condition is satisfied. In a case where the transmission condition is not satisfied, the process returns to Step **S10**. In a case where the target apparatus **12** restarts the operation, the processes of Steps **S10** to **S16** are repeated. As a result, a plurality of pieces of partial operation sound data **42** may be stored in the volatile memory **38**. In a case of acquiring the partial operation sound data **42**, the transmission processing unit **54** may immediately transmit the partial operation sound data **42** to the data checker.

(71) In Step **S18**, the transmission processing unit **54** determines whether or not the partial operation sound data **42** is stored in the volatile memory **38**. In a case where the transmission condition is that the time has reached a predetermined time point, and the target apparatus **12** has never operated during a period to the predetermined time point, that is, in a case where the partial operation sound data **42** is not stored in the volatile memory **38**, the process is ended. In a case where the partial operation sound data **42** is stored in the volatile memory **38**, the process proceeds to Step **S20**.

(72) In Step **S20**, the transmission processing unit **54** adds a priority to each piece of partial operation sound data **42** stored in the volatile memory **38** based on at least one of the sound volume of the abnormal sound **N** contained in the partial operation sound data **42**, the data volume of the partial operation sound data **42**, or the occurrence time point of the abnormal sound **N**.

(73) In Step **S22**, the transmission processing unit **54** transmits a plurality of pieces of partial operation sound data **42** with the priority to the data checker. Then, the data checker checks

whether or not the partial operation sound data **42** contains the confidential information of the organization in the order indicated by the priority, and transmits the partial operation sound data **42** checked not to contain the confidential information, to the out-of-organization processing apparatus **16**.

(74) FIG. **8** is a flowchart illustrating a processing flow of the target apparatus **12** in the exemplary embodiment in which the transmission processing unit **54** directly transmits the partial operation sound data **42** to the out-of-organization processing apparatus **16**. Processes of Steps **S40** to **S50** in the flowchart illustrated in FIG. **8** are identical to the processes of Steps **S10** to **S20** in the flowchart illustrated in FIG. **7**. Thus, the repetitive description will be omitted here.

(75) In Step **S52**, the transmission processing unit **54** transmits, to the out-of-organization processing apparatus **16**, the partial operation sound data **42** selected based on the priority added to each piece of partial operation sound data **42** in Step **S50** among a plurality of pieces of partial operation sound data **42** stored in the volatile memory **38**. Typically, the transmission processing unit **54** transmits only the partial operation sound data **42** having the highest priority to the out-of-organization processing apparatus **16**.

(76) Hitherto, the exemplary embodiment of the present invention has been described, but the exemplary embodiment of the present invention is not limited to the above exemplary embodiment, and various modifications may be made without departing from the spirit of the present invention.

(77) For example, in the above exemplary embodiment, the target apparatus **12** is an information processing apparatus, and the target apparatus **12** has the functions as the operation sound data acquisition processing unit **48**, the abnormal sound detection unit **50**, the partial operation sound data acquisition unit **52**, and the transmission processing unit **54**. Another apparatus other than the target apparatus **12** may have such functions. For example, an in-organization server that is communicably connected to the target apparatus **12** and is managed in the organization may have the functions as the operation sound data acquisition processing unit **48**, the abnormal sound detection unit **50**, the partial operation sound data acquisition unit **52**, and the transmission processing unit **54**. In this case, the in-organization server corresponds to the information processing apparatus.

(78) The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

Claims

1. An information processing apparatus comprising: a processor configured to: acquire operation sound data being data obtained by recording an operation sound of a target apparatus used in an organization; detect an abnormal sound from the operation sound data; and transmit partial operation sound data to an out-of-organization processing apparatus, and not transmit a portion of the operation sound data other than the partial operation sound data to the out-of-organization processing apparatus, the partial operation sound data being data that contains the abnormal sound and is a portion of the operation sound data, and the out-of-organization processing apparatus being an apparatus that is managed outside the organization and executes a process related to the abnormal sound.

2. The information processing apparatus according to claim 1, wherein the processor is configured to: transmit the partial sound data to a data checker who transmits the partial operation sound data

to the out-of-organization processing apparatus after checking the partial operation sound data, and transmit the partial operation sound data to the out-of-organization processing apparatus in response to a notification from the data checker who has checked that confidential information of the organization is not contained in the partial operation sound data.

3. The information processing apparatus according to claim 1, wherein the processor is configured to: acquire a plurality of pieces of the operation sound data and acquire a plurality of pieces of the partial operation sound data, each being a portion of each piece of the operation sound data, and transmit, to the out-of-organization processing apparatus, the partial operation sound data selected based on at least one of a sound volume of the abnormal sound, a data volume of the partial operation sound data, or an occurrence time point of the abnormal sound, among the plurality of pieces of the partial operation sound data.

4. The information processing apparatus according to claim 1, wherein a time length of the partial operation sound data is a time required to specify a cause of the abnormal sound.

5. The information processing apparatus according to claim 1, wherein a time length of the partial operation sound data is a time corresponding to a type of the abnormal sound.

6. A non-transitory computer readable medium storing an information processing program causing a computer to execute a process comprising: acquiring operation sound data being data obtained by recording an operation sound of a target apparatus used in an organization; detecting an abnormal sound from the operation sound data; and transmitting partial operation sound data to an out-of-organization processing apparatus, and not transmitting a portion of the operation sound data other than the partial operation sound data to the out-of-organization processing apparatus, the partial operation sound data being data that contains the abnormal sound and is a portion of the operation sound data, and the out-of-organization processing apparatus being an apparatus that is managed outside the organization and executes a process related to the abnormal sound.

7. An information processing apparatus comprising: acquiring operation sound data being data obtained by recording an operation sound of a target apparatus used in an organization; detecting an abnormal sound from the operation sound data; and transmitting partial operation sound data to an out-of-organization processing apparatus, and not transmit a portion of the operation sound data other than the partial operation sound data to the out-of-organization processing apparatus, the partial operation sound data being data that contains the abnormal sound and is a portion of the operation sound data, and the out-of-organization processing apparatus being an apparatus that is managed outside the organization and executes a process related to the abnormal sound.
